

Valliammai Valliappan

Software Engineer

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SUMMARY

Passionate and results-driven **Software Engineer** specialized in **microservices development and containerization**, with hands-on experience in **Golang, Kubernetes, and Docker**. While working at **Dell Technologies**, I have contributed to building and enhancing catalog generation service architectures to improve efficiency and scalability. With a strong academic background as a **Merit Scholar in Computer Science**, I bring innovative problem-solving skills, a strong work ethic, and exceptional organizational abilities. In addition to backend expertise, I have experience in **deep learning, machine learning**, and **data visualization tools like Tableau**, enabling data driven decision-making.

EDUCATION

Vellore Institute of Technology

Master of Technology

Major in Computer Science Specialisation in Data science;

Cumulative GPA: 9.69/10

Vellore, India

June 2019 - Aug 2024

WORK EXPERIENCE

Dell Technologies

Software Engineer 2

Bangalore, India

Jun 2024 - May 2025

- **Automated BSpec Breakdown into epics and user stories with AI Agents** was authorised and patent by Dell/DellEMC Invention Disclosure
- **Optimized Whole Update Package Aggregation in Dell UpdatePackage Publisher** by implementing concurrency techniques, such as Go routines, reduced processing time from over 2 days to just 9 hours through which **81.25%** of performance enhancement is achieved.
- **Adapted the Dell UpdatePackage Publisher User Interface** to align with the Dell Design System by learning React and implementing the required changes within 2 weeks, improving UI consistency. And reduced 20% of waiting time in creation of profile management.
- **Developed a Chatbot for Dell Update Package Creation Services (DCS)** using Llama 3.2:3b and FIASS to handle repetitive customer queries, enhancing customer service efficiency. It reduced 50% of customer queries to be handled by the development team.
- **Built an Event Notification System** that aggregates new update packages within seconds, ensuring real-time updates. This involves kafka notification and http share watcher.
- **Designed an Altered Package Aggregator** to generate package XML for legacy Dell Update Package Creation (DUP), streamlining the packaging process.
- **Implemented Feature Flags** by deploying Flagmaith and integrating it into the enterprise catalog generator application, enabling dynamic feature control.
- **Tech Stack:** Golang, Microservice, Kubernetes, LLM, AI Agents, Feature Flags, React, Kafka, Grafana, LangChain, Distributed System, httpshare watcher,Postgresql, Python.

Dell Technologies

Graduate Intern

Bangalore, India

July 2023 – May 2024

- Contributed to the transition of Catalog generation service from **monolithic services to microservices**, enhancing scalability and maintainability. Specialized in containerization, improving application deployment and management using **Docker and Kubernetes**.
- Automated the transformation of **XSD files into POST API calls**, reducing manual workload and errors.
- **Developed an Error Audit Microservice** that captures errors occurring within microservices and sends real-time alerts to a **Teams webhook**, improving issue tracking and response time.
- **Tech Stack:** Golang, Microservice, Kubernetes, MongoDB, Automation, Logstash, Elastic search, Devops .

UNIVERSITY PROJECTS

Palm Leaf Letter Detection and Recognition

2023

- Aims to digitize and decipher ancient palm leaf texts.
- Developed a project for palm leaf manuscript detection and recognition, achieving a 72% accuracy rate, utilizing techniques like CRAFT and VGGNet.
- **Tech Stack** : Python, Pytorch, VGGNet models, CRAFT

Decentralised Music App

2023

- The aim is to allow musicians and distributors to share details of rights and royalties for music tracks online. These smart contracts automatically license recordings and take payment each time they are played.
- **Tech Stack** : Node.js, Solidity, Ganache, Truffle, Blockchain

Memory Assistance

2022

- Utilizes advanced deep learning techniques for therapeutic applications
- Implemented a facial recognition project to assist patients with amnesia, incorporating interactive games and music therapy
- Achieved an 80% reduction in training time using Siamese network architecture
- **Tech Stack** : Python, Tensorflow, CNN models, Siamese Network, OpenCV, SQL

Farm easy

2022

- A website for farmers to share their machinery and sell their products directly.
- **Tech Stack** : Node.js, MongoDB, Agile methodology, Requirement gathering

ADDITIONAL

Technical Skills: Kubernetes, Docker, Golang, C++, Deep Learning, MongoDB, NLP, Python, REST API, Streamlit, Tableau, React.js, Node.js, Microservices, LLM, Java, Kafka, AWS S3, AWS EC2, Devops, NoSQL, CI/CD pipelines.

Languages: Fluent in English, Tamil;

Awards: Merit scholar throughout all years of study; first place in speedocode contest 2022;